AMENDMENT NO. 1 DECEMBER 2016
TO
IS 2113 : 2014 ASSAYING OF SILVER IN SILVER AND SILVER ALLOYS — METHODS

(Third Revision)

(*Page* 2, *clause* **6.5.1**, *lines*1 *and* 2) — Substitute 'Weigh in triplicate of 1.00 g of each sample accurately upto 0.01 mg' *for* 'Weigh two number of 1.00 g of each sample accurately up to 0.01mg'.

(*Page* 2, *clause* **6.7**, *last sentence*) — Insert the following lines before the last sentence:

'One (outlier) determination may be discarded leaving only duplicate determination. If those are within 0.5 ppt (0.2 for fine silver) then the requirement of duplicate determination is satisfied.'

(MTD 10)

Publication Unit, BIS, New Delhi, India

भारतीय मानक
Indian Standard

IS 2113 : 2014
(Reaffirmed 2019)

चाँदी एवं चाँदी मिश्रधातुओं में चाँदी की परख — पद्धतियाँ

(तीसरा पुनरीक्षण)

Assaying Silver in Silver and Silver Alloys — Methods

(Third Revision)

ICS 77.120.01, 39.060

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December 2014 **Price Group 2**

Precious Metals Sectional Committee, MTD 10

FOREWORD

This Indian Standard (Third Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Precious Metals Sectional Committee had been approved by the Metallurgical Engineering Division Council.

The methods are intended to determine fineness of silver in silver and silver alloys as covered in IS 2112 : 2003 'Silver and silver alloys, jewellery/artefacts – Fineness and marking – Specification' except for fineness of 999.9.

This standard has been aligned with ISO 13756 : 1997 for volumetric method. The revision incorporate's Amendment No. 1 (July 2008), Amendment No. 2 (June 2010) and Amendment No. 3 (December 2010).

In this revision 'scope' in clause **1**, 'NOTE' in clause **7.3.4** and 'repeatability' in clause **10** has been modified. Clause **7** to clause **11** has been renumbered true to the omission of clause **6** 'NOTE' of clause **8.2.1** has been deleted.

For the purpose of deciding whether particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

ASSAYING SILVER IN SILVER AND SILVER ALLOYS — METHODS

(Third Revision)

1 SCOPE

This standard prescribes the methods for determination of silver in silver alloys as below:

a) Gravimetric method, and
b) Volumetric (potentiometric) method.

The methods are suitable for determination of silver in the ranges as specified in IS 2112 except for fineness 999.9 for which the test method may be agreed to between the parties concerned since there associated impurities have to be determined. The silver alloys may contain copper (Cu), zinc (Zn), cadmium (Cd) and palladium (Pd). These elements except palladium, which must be precipitated before titration, do not interfere with this method of determination of silver.

The volumetric method can be used for silver of fineness 99.5 percent and below. These alloys may contain copper, zinc, cadmium and palladium. These elements, except palladium, which must be precipitated before titration, do not interfere with this method of determination of silver. In case of dispute, Gravimetric method may be used as the reference method.

2 REFERENCES

The standards listed below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below.

IS No.	*Title*
264 : 1976	Nitric acid (*second revision*)
265 : 1993	Hydrochloric acid (*fourth revision*)
1070 : 1992	Reagent grade water (*third revision*)
2112 : 2002	Silver and silver alloys, jewellery/ artefacts — Fineness and marking — Specification (*third revision*)

3 TERMINOLOGY

For purpose of the standard, the following definitions shall apply.

3.1 High precision balance capable of weighing to an accuracy of 0.01 mg can be used. For calculation of silver in parts per thousand procedure given at **6.6** shall be used. While using such balances the weight of samples shall be 1 g for gravimetric methods and 0.5 g for volumetric (potentiometric) method respectively.

3.2 Check Silver

Silver of known fineness but not less than 999 for jewellery articles or artefacts but for estimation of fine silver check silver shall be of 999.9 fineness.

4 SAMPLING

4.1 Dip Sampling

A dip sample from the representative lot shall be taken with a sampler from the molten metal just before casting and granulated or cast into a button. This may be hammered, rolled, cleaned and cut into small pieces for assaying.

4.2 Two samples shall be taken by cutting or drilling diagonally opposite corners from the upper and lower sides if the material is in the form of ingot, slab or bar. The samples shall then be cut into small pieces and thoroughly mixed together.

NOTE — Iron particles, if any, may be separated with the help of a magnate before weighment of the sample.

4.3 When the material is in the form of sheets or wire, small pieces shall be cut preferably from opposite ends and thoroughly mixed together.

4.4 In case of dispute with regard to sampling of ingots, slab, bar, sheet or wire, a dip sample shall be taken from the representative lot.

4.5 Before taking samples by any of the above methods, all surface dirt from the article shall be removed and all instruments for cutting, scrapping, hammering et c, shall be cleaned thoroughly. No oil shall be used for drilling.

4.6 The surface of the bar shall be well scrapped before sampling or the first portion of metal removed during the preparation of the sample shall be rejected if the bar has been pickled by dipping in acid after casting.

4.7 For coated articles, appropriate precautions as agreed shall be taken to ensure exclusion of coating.

5 QUALITY OF REAGENTS

5.1 Unless specified otherwise, analytical grade chemical and reagent grade water shall be employed for the tests.

6 GRAVIMETRIC METHOD

6.1 Sampling

As given in **4.1** to **4.7**

6.2 Outline of the Method

The sample is dissolved in nitric acid and silver is precipitated as silver chloride, filtered through previously weighed sintered glass crucible, wash, dried and weighed.

6.3 Reagents

6.3.1 *Dilute Nitric Acid* —1:1 and 3:97 (*v/v*)

6.3.2 *Dilute Hydrochloric Acid* —1:9 (*v/v*)

6.4 Apparatus

6.4.1 *Balance of Accuracy* — 0.01 mg.

6.4.2 *250 ml Glass Beakers*

6.4.3 *75 mm Dia Watch Glasses*

6.4.4 *Glass Rods Grounded at Both the Ends*

6.4.5 *Sintered Glass Crucibles No. 3* — medium porosity.

6.4.6 *Wash Bottle with Fine Jet*

6.4.7 *Filter Vacuum Pump*

6.4.8 *Drying Oven (Electrical) Up to 250°C*

6.4.9 *Desiccator*

6.5 Procedure

6.5.1 Weigh two number of 1.00 g of each sample accurately up to 0.01 mg. Transfer into three number of 250 ml beakers. Add 15 ml of 1:1 nitric acid, cover the beakers with watch glass and heat gently on the hot plate till the samples are completely dissolved. Expel all the nitrous fumes.

6.5.2 Remove the beakers from the hot plate and remove the watch glass by washing with a fine jet avoiding spurting. Dilute the content to about 100 ml with distilled water. Add 20 ml of hydrochloric acid to it slowly with continuous stirring avoiding any loss of silver chloride while stirring. Cover the beakers with watch glass and keep the beakers on hot plate, avoiding boiling. After one hour check with few drops of dilute hydrochloric for the complete precipitation of silver. Keep on the hot plate at least for 2 h (preferably over night in a dark place).

6.5.3 Remove the watch glass from the beakers while washing very gently with a fine jet of reagent grade water, wash down the walls of the beaker with fine jet, and filter through previously cleaned, dried and weighed sintered glass crucible using vacuum pump (*see* **6.4.7**). Adjust the rate of filtration, by adjusting the pressure of the pump so that it is neither too fast nor too slow. Wash the precipitate 2-3 times with dilute nitric acid (3 percent) and then 5 to 6 times with distilled water.

6.5.4 Dry the sintered glass crucibles with silver chloride precipitate in electric oven (*see* **6.4.8**) at 140°C for at least 1 h. Cool the crucibles in desiccator (*see* **6.4.9**) to room temperature and weight after one hour. Repeate the procedure till constant weight is obtained.

6.6 Calculations

$$\text{Silver, part per thousand} = \frac{A \times 0.7526 \times 10^{3}}{B}$$

where

A = mass of silver chloride, in g; and

B = mass of the sample, in g.

NOTES

1 After dissolution of the samples (*see* **6.5.2**) in nitric acid if gold is present, filter through a Whattman filter paper No.1 and wash the filter paper containing gold several times with water till washing is free from silver.

2 In case tin is present, evaporate the contents on low heat to syrupy consistency to precipitate tin as meta stannic acid. Add 5 ml of nitric acid and 20 ml water and allow the beaker to stand on hot plate to settle down the precipitate. Filter the precipitate through Whattman filter paper No. 42, wash the precipitate at least five-six times with nitric acid till the washing is free from silver. Continue the precipitation of silver with hydrochloric acid as given in **6.5.2.**

6.7 Repeatability

Duplicate determinations shall give results differing by not more than 0.5 parts per thousand(‰) respectively by mass for silver alloys and not more than 0.2 parts per thousand (‰) respectively for fine silver. When the difference is greater than this assay shall be repeated.

7 DETERMINATION OF SILVER IN SILVER ALLOYS BY VOLULMETRIC (POTENTIOMETRIC) METHOD USING SODIUM CHLORIDE OR POTASSIUM CHLORIDE

7.1 Principle

The sample is dissolved in dilute nitric acid. The silver content of the resulting solution is determined by titration with Standard Sodium Chloride or Potassium Chloride Solution using potentiometric indication of the equivalence point.

7.2 Reagents

7.2.1 *Nitric Acid* — 33 percent (*v/v*), sp. gr. 1.200, free of Halide.

7.2.2 *Sodium Chloride Solution* — 0.25 *mol/litre.*

Dissolve 14.6 g of sodium chloride (dried at 105°C) in water and dilute to 1 000 ml.

7.2.3 *Potassium Chloride Solution* — 0.25 *mol/litre*

Dissolve 18.6 g of potassium chloride (dried at 105°C) in water and dilute to 1 000 ml.

7.2.4 *Check Disodium Dimethylglyoxime Solution*

Dissolve 10 g of disodium dimethylglyoxime octahydrate in 1 000 ml of water.

7.2.5 *Silver*

7.2.5.1 For the determination of silver alloys, silver jewellery/artefacts of purity 990 parts per thousand and below, fine silver of purity 999 parts per thousands shall be used.

7.2.5.2 For the determination of silver in silver alloys, silver jewellery/artefacts, purity above 990 parts per thousand, fine silver of purity 9999 parts per thousand shall be used.

7.3 Apparatus

7.3.1 *Ordinary Laboratory Glassware and Apparatus*

7.3.2 Motor driven plunger or piston-type burette linked to potentiometer, preferably an automatic titrator capable of delivering increment of 0.05 ml at the equivalence point.

7.3.3 Titration apparatus with combination silver electrode or silver chloride coated silver electrode and mercury/mercury sulphate (Hg/Hg_2SO_4) electrode or other suitable reference electrode.

7.3.4 Titration beaker with automatic titrator.

NOTE — It is preferable to use automatic titrator to get accurate and reproducible result within limit of tolerance of 1 ppt.

8 PROCEDURE

8.1 Test Portion

The mass of sample used for the titration shall contain between 300 mg and 500 mg of silver and shall be weighed to an accuracy of 0.01 mg.

8.2 Determination of Sodium Chloride or Potassium Chloride Standard Solution Factor

8.2.1 *Preparation of Silver Standards*

Weigh to the nearest 0.01 mg, three samples, each of 300 mg to 500 mg of silver and transfer them to glass beakers. Add 5 ml of nitric acid (*see* **7.2.1**) to each, and warm gently (preferably in water bath) to dissolve the silver. Keep the tops of the beakers covered with watch glasses. Heat until evolution of nitrogen oxides ceases. Allow it to cool. Rinse the watch glasses into beakers. Transfer to the titration beaker (*see* **7.3.4**). Add the minimum volume of water, with in the range of about 20 ml to 60 ml to satisfy the requirements of the titration apparatus (*see* **7.3.3**) with respect to measurement and stirring.

8.2.2 *Titration of Standard Silver Solution*

Add via the plunger-burette and with continuous stirring, sufficient sodium chloride (*see* **7.2.2**) or potassium chloride (*see* **7.2.3**) to precipitate about 90 percent of silver in silver solution. Titrate the remaining silver in such a manner that the equivalence point can be interpolated from 0.05 ml additions of sodium chloride solution

NOTE — This split titration approach may be effected automatically using an automatic titrator with so called dynamic volume dosing based on the measured potential difference across the electrodes in the titration vessels (*see* **7.3.3**).

8.2.3 *Calculation of Sodium Chloride or Potassium Chloride factor*

The sodium chloride or potassium chloride factor *F*, is calculated using the formula :

$$F = \frac{m_{Ag}}{V_s}$$

where

m_{Ag} = mass of silver, in mg; and

V_s = volume in ml of sodium chloride or potassium chloride solution at the equivalence point.

The factor determinations should not differ from each other by more than 0.05 parts per hundred (%) by mass relative value.

The mean value *F*, shall be used in subsequent calculations for maximum accuracy. The sodium chloride or potassium chloride factor shall be determined immediately before analysis of the test samples.

8.3 Determination

8.3.1 *Preparation of Test Solutions*

Weigh two test samples each containing between 300 mg to 500 mg silver to an accuracy of 0.01 mg and transfer into glass beaker or titration beaker with automatic titrator. Add 5 ml of nitric acid (*see* **7.2.1**). Keep the beakers covered with watch glasses. Heat until evolution of nitrogen oxides ceases. Allow to cool.

Rinse the water glasses into the respective beakers. Transfer the solution to titration apparatus (*see* **7.3.3**). Add water as per **8.2.1**.

NOTE— When Teflon beaker supplied with automatic titrator is used along with hot water bath for dissolution of samples, it can be directly fixed in the titration apparatus.

8.3.2 *Elimination of Palladium*

Palladium shall be eliminated by addition of an aqueous solution of disodium dimethylglyoxime octahydrate (*see* **7.2.3**). For each 100 mg palladium add 20 ml of this solution and stir before commencing titration.

8.3.3 *Titration of Test Solution*

Proceed exactly as per the standard solution. It may be necessary to carryout a pilot determination to obtain an approximate value of the silver content.

9 EXPRESSION OF RESULTS

9.1 Method of Calculation

Since the sodium chloride or potassium chloride factor F (*see* **8.2**) is expressed in mg of silver for each ml of solution, the mass m_{Ag}, in mg, of silver in the test portion is calculated using the following formula:

$$m_{Ag} = F \times V_s$$

where

F = sodium chloride or potassium chloride factor, and

V_s = volume in ml of sodium chloride or potassium chloride solution at the ml equivalence point

9.2 Calculate the silver content of the samples W_{Ag}, in parts by mass per thousand (‰), using the formula:

$$W_{Ag} = \frac{m_{Ag}}{m_s} \times 1\,000$$

where

m_s = mass in mg, of the test portion (*see* **8.1**); and

m_{Ag} = mass of silver in mg.

10 REPEATABILITY

The results of duplicate determinations shall correspond to better than 1 part by mass per thousand (‰) of silver. If the variation is greater than this, the assay shall be repeated.

Bureau of Indian Standards

BIS is a statutory institution established under the *Bureau of Indian Standards Act*, 1986 to promote harmonious development of the activities of standardization, marking and quality certification of goods and attending to connected matters in the country.

Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards : Monthly Additions'.

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Amendments Issued Since Publication

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